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Case 37-2022: A 55-Year-Old Man with Fatigue, Weight Loss, and Pulmonary Nodules

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PRESENTATION OF CASE

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Dr. Jarone Lee: A 55-year-old man was admitted to this hospital 7 months after kidney transplantation because of fatigue, weight loss, and new pulmonary nodules.

The patient had been in his usual state of health until 1 week before this admission, when severe fatigue and generalized weakness developed. He had lost 4.5 kg in the past month after making healthier dietary choices; however, he had also noticed abdominal discomfort and a decrease in appetite.

During the next week, the patient was able to eat and drink very little and lost an additional 4.5 kg. The fatigue and weakness worsened, and he mostly stayed in bed. He had several episodes of lightheadedness, gait instability, and falls while he was walking to the bathroom. There was new odynophagia, dysphagia, and nausea.

On the day of admission, the patient was evaluated at the transplant nephrology clinic of this hospital before a scheduled infusion of belatacept. The temperature was 37.3°C, the blood pressure 70/50 mm Hg, the heart rate 98 beats per minute, the respiratory rate 35 breaths per minute, and the oxygen saturation 96% while he was breathing ambient air. He appeared cachectic and lethargic. He was transported by ambulance from the clinic to the emergency department of this hospital.

In the emergency department, the patient reported malaise and feeling like he had no energy or strength. The lightheadedness and gait instability persisted. A review of systems was notable for shortness of breath, dark urine, and continued anorexia, nausea, odynophagia, dysphagia, and abdominal discomfort. He reported no chills, night sweats, cough, chest pain, vomiting, hematochezia, melena, or dysuria.

The patient had a history of sarcoidosis. Nine years before this admission, nephrocalcinosis caused end-stage kidney disease. Hemodialysis was started, and this treatment was continued until a deceased-donor kidney transplantation was performed 7 months before this admission. Routine serologic testing performed before transplantation was positive for Epstein-Barr virus (EBV) IgG and cytomegalovirus (CMV) IgG. An interferon- γ release assay for *Mycobacterium tuberculosis* was

negative. Serologic testing in the donor was also positive for EBV IgG but was negative for CMV IgG. Induction immunosuppressive therapy with antithymocyte globulin was initiated; maintenance therapy included prednisone, mycophenolate mofetil, and tacrolimus.

Six months before the current admission, pathological examination of a biopsy specimen from the transplanted kidney revealed vascular disease of donor origin and acute tubular injury but no evidence of T-cell-mediated or antibody-mediated rejection. Treatment with tacrolimus was stopped and belatacept started; prednisone and mycophenolate mofetil therapy was continued.

One month before the current admission, pathological examination of another biopsy specimen from the transplanted kidney revealed focal infiltrates that were vaguely granulomatous and were associated with ruptured tubules and interstitial Tamm-Horsfall protein (also known as uromodulin). There was no evidence of allograft rejection.

Two weeks before the current admission, laboratory tests revealed a blood creatinine level of 2.31 mg per deciliter (204.2 μ mol per liter; reference range, 0.60 to 1.50 mg per deciliter [53.0 to 132.6 μ mol per liter]); routine laboratory tests had revealed similar creatinine levels during the previous 6 months. Other laboratory test results are shown in Table 1.

The patient also had a history of hypertension, hyperlipidemia, and gout. Current medications included aspirin, atorvastatin, labetalol, nifedipine, trimethoprim-sulfamethoxazole, valganciclovir, prednisone, mycophenolate mofetil, and belatacept. There were no known drug allergies. The patient lived with his mother in an urban area of New England and had never traveled outside the region. He worked as an administrator and had never been homeless or incarcerated. He had no sexual partners and did not smoke tobacco, use illicit drugs, or drink alcohol.

On the day of evaluation in the emergency department, the temperature was 36.7°C, the blood pressure 80/50 mm Hg, the heart rate 100 beats per minute, the respiratory rate 24 breaths per minute, and the oxygen saturation 92% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 20.5. The patient was lethargic and spoke in sentences of three or four words. The mucous membranes were

dry, and the throat could not be evaluated because of nausea. There was no cervical lymphadenopathy. Auscultation of the lungs revealed diffuse inspiratory crackles. Neurologic examination was limited but was notable for 4/5 motor strength in the arms and legs.

The blood level of creatinine was 5.05 mg per deciliter (446.4 μ mol per liter), the calcium level 13.1 mg per deciliter (3.3 mmol per liter; reference range, 8.5 to 10.5 mg per deciliter [2.1 to 2.6 mmol per liter]), the lactic acid level 4.4 mmol per liter (39.6 mg per deciliter; reference range, 0.5 to 2.0 mmol per liter [4.5 to 18.0 mg per deciliter]), and the hemoglobin level 6.6 g per deciliter (reference range, 13.5 to 17.5). Cultures of blood were obtained. Other laboratory test results are shown in Table 1.

Dr. Mark C. Murphy: Computed tomography (CT) of the chest, abdomen, and pelvis was performed without the administration of intravenous contrast material. CT of the chest (Fig. 1) revealed innumerable bilateral miliary pulmonary nodules that were new relative to a CT scan that had been obtained 6 months earlier. The nodules were in a random distribution that was suggestive of a hematogenous origin. Trace bilateral pleural effusions were present, as was calcified mediastinal and bilateral hilar lymphadenopathy; the lymphadenopathy appeared unchanged from previous imaging. CT of the spleen (Fig. 2) revealed new splenomegaly. There was new mild dilatation of the renal collecting system of the transplanted kidney in the right lower quadrant of the abdomen.

Dr. Lee: While the patient was being evaluated in the emergency department, the temperature increased to 39.6°C. Intravenous fluids and intravenous infusion of phenylephrine were administered. Empirical treatment with vancomycin, cefepime, metronidazole, levofloxacin, doxycycline, and micafungin was started; trimethoprim-sulfamethoxazole and valganciclovir were continued. Treatment with prednisone and mycophenolate mofetil was discontinued, and hydrocortisone therapy was started. The patient was admitted to the intensive care unit.

Within 24 hours after admission, the oxygen saturation had decreased to 84% while the patient was breathing ambient air; supplemental oxygen was administered through a nasal cannula at a rate of 2 liters per minute, and the oxygen saturation increased to 94%. Continuous intra-

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	2 Wk before Admission	On Admission
Blood			
White-cell count (per μ l)	4500–11,000	7490	3570
Differential count (per μ l)			
Neutrophils	1800–7700	6880	3380
Lymphocytes	100–4800	190	120
Monocytes	200–1200	300	60
Eosinophils	0–900	10	—
Basophils	0–300	20	—
Sodium (mmol/liter)	135–145	133	132
Potassium (mmol/liter)	3.4–5.0	4.5	5.4
Chloride (mmol/liter)	98–108	93	95
Carbon dioxide (mmol/liter)	23–32	17	15
Urea nitrogen (mg/dl)	8–25	33	95
Creatinine (mg/dl)	0.60–1.50	2.31	5.05
Glucose (mg/dl)	74–106	111	102
Calcium (mg/dl)	8.5–10.5	10.1	13.1
Phosphorus (mg/dl)	2.6–4.5	3.0	5.1
Lactic acid (mmol/liter)	0.5–2.0	—	4.4
Alanine aminotransferase (U/liter)	10–55	—	44
Aspartate aminotransferase (U/liter)	10–40	—	79
Alkaline phosphatase (U/liter)	30–100	—	171
Total bilirubin (mg/dl)	0–1.0	—	0.6
Globulin (g/dl)	1.9–4.1	—	3.3
Albumin (g/dl)	3.3–5.0	3.5	2.8
Hemoglobin (g/dl)	13.5–17.5	7.8	6.6
Hematocrit (%)	41.0–53.0	26.0	23.6
Platelet count (per μ l)	150,000–400,000	185,000	144,000
Prothrombin time (sec)	11.5–14.5	15.0	15.8
International normalized ratio	0.9–1.1	1.2	1.3
Activated partial-thromboplastin time (sec)	22.0–35.0	54.4	—
Urine			
Bilirubin	Negative	—	Negative
Urobilinogen	Negative	—	Negative
Blood	Negative	—	1+
Glucose	Negative	—	Negative
Ketones	Negative	—	Negative
Leukocyte esterase	Negative	—	Negative
Nitrite	Negative	—	Negative

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

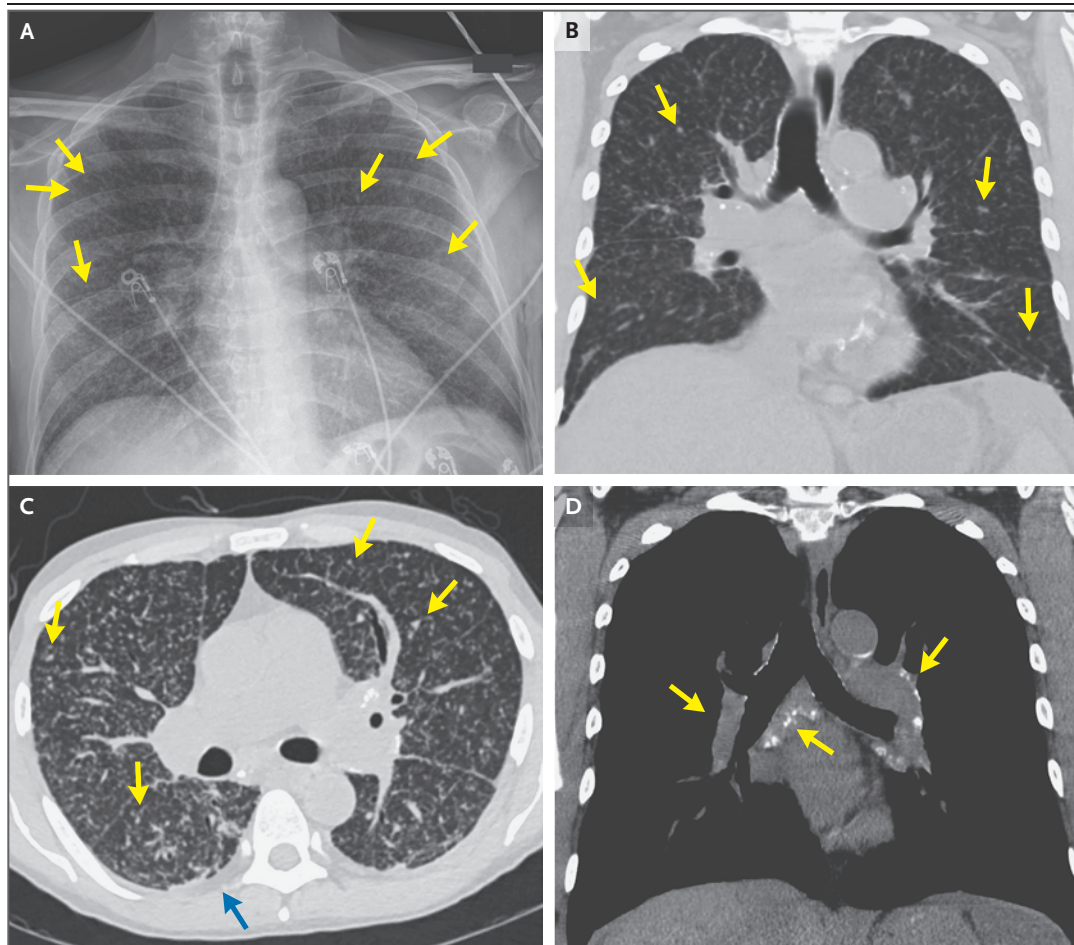


Figure 1. Imaging Studies of the Chest Obtained on Admission.

A frontal radiograph (Panel A) shows multiple tiny pulmonary nodules (arrows) scattered diffusely throughout both lungs in a miliary pattern. Coronal (Panel B) and axial (Panel C) images from a CT on lung windows show numerous small bilateral nodules (yellow arrows) that are new relative to imaging obtained 6 months earlier. The distribution of nodules is random, a characteristic that is suggestive of a hematogenous origin. There is a small right pleural effusion (Panel C, blue arrow). A coronal CT image obtained with soft-tissue windows (Panel D) shows enlarged bilateral hilar and subcarinal lymph nodes that are partially calcified (arrows). These findings are consistent with the known history of chronic sarcoidosis.

venous infusion of phenylephrine was continued, and norepinephrine was added to maintain a mean arterial blood pressure above 65 mm Hg. Two units of packed red cells were transfused. The creatinine level decreased to 3.82 mg per deciliter (337.7 μ mol per liter), the lactic acid level to 1.6 mmol per liter (14.4 mg per deciliter), and the calcium level to 8.9 mg per deciliter (2.2 mmol per liter). Treatment with levofloxacin and micafungin was stopped; isavuconazole was started.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Matthew B. Roberts: This 55-year-old man with a history of sarcoidosis and end-stage kidney disease, for which he underwent kidney transplantation, presented with a progressive and fulminant clinical syndrome that is notable for fever, pulmonary nodules, splenomegaly, and hypercalcemia — findings that will form the basis of my differential diagnosis. Because this patient is a kidney-transplant recipient, I will also construct a differential diagnosis in the context of his net

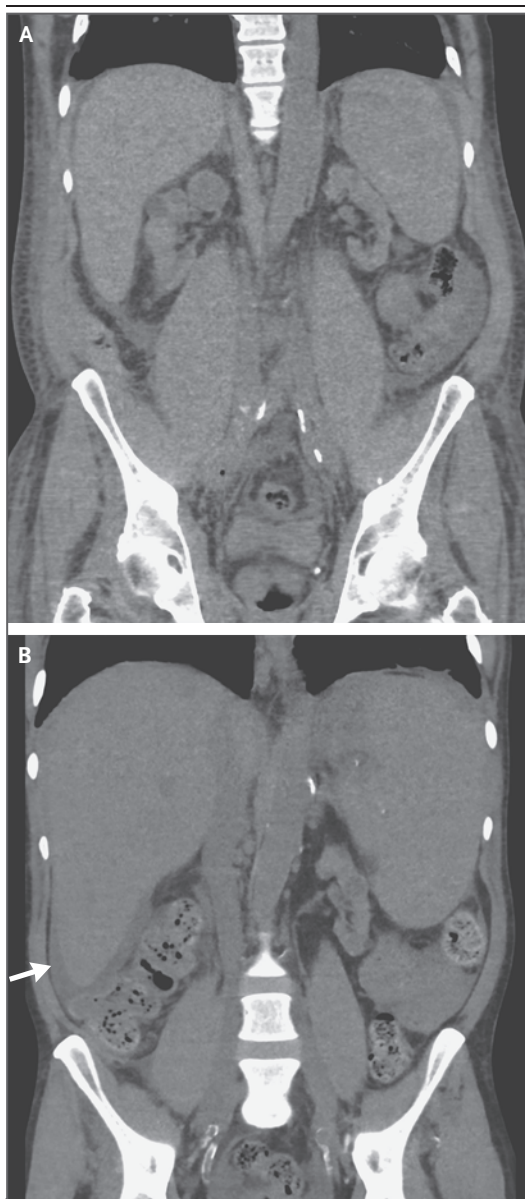


Figure 2. CT of the Abdomen and Spleen.

Shown are coronal images of the abdomen obtained with soft-tissue windows from CT performed without the administration of intravenous contrast material 6 months before admission (i.e., baseline) (Panel A) and on admission (Panel B). There has been development of splenomegaly, with the spleen measuring 16 cm on admission. A small volume of new perihepatic ascites is also present (arrow).

state of immune suppression and the described timeline of infections after solid-organ transplantation.^{1,2} Given that pulmonary nodules are a prominent finding in this case, I will discuss potential causes of pulmonary nodules before

narrowing the differential diagnosis by considering the patient's other pertinent features — splenomegaly and hypercalcemia.

PULMONARY NODULES

The development of pulmonary nodules is a relatively common problem after solid-organ transplantation and frequently prompts an infectious diseases consultation. Pulmonary nodules are often the first feature identified in many unwell immunosuppressed patients, including solid-organ transplant recipients, and the lungs are usually the most accessible site for obtaining diagnostic samples.

Unfortunately, pulmonary nodules can have many causes, and radiologic features can be non-specific.³ This patient's presentation includes fever, which necessitates consideration of infectious causes of pulmonary nodules. Common bacterial infections are unlikely to cause pulmonary nodules and rarely manifest with discrete small nodules, but uncommon pathogens such as legionella or nocardia may produce this radiologic appearance. This patient is receiving trimethoprim-sulfamethoxazole prophylaxis, which makes nocardia a relatively unlikely cause, although breakthrough infection despite prophylaxis can occur.⁴

Bacterial endocarditis on the right side of the heart can cause septic pulmonary emboli; however, this patient has no risk factors for endocarditis, no suggestive clinical signs such as a murmur, and no evidence of peripheral emboli. Furthermore, the large number of pulmonary nodules is not consistent with a diagnosis of endocarditis. Mycobacterial infections, including infections with *M. tuberculosis* and nontuberculous mycobacteria, often cause pulmonary nodules that are consistent with the clinical and radiologic features observed in this patient.⁵

Fungal infections are important opportunistic infections to consider in transplant recipients.¹ This patient underwent kidney transplantation 7 months before this admission and is probably having increasing environmental exposures as he experiences life without dialysis, meaning that he is at high risk for fungal infection.¹ Fungal infections frequently cause pulmonary nodules; commonly encountered organisms include aspergillus species, mucorales, and endemic fungi such as *Histoplasma capsulatum*, *Blastomyces dermatitidis*, and *coccidioides*. It is important to consider *Pneumocystis jirovecii* infection in any immunocompromised

host with pulmonary symptoms; however, this is an unlikely diagnosis in a patient who is receiving trimethoprim–sulfamethoxazole prophylaxis, which is highly effective in preventing pneumocystis pneumonia.⁶

Viral infections that warrant consideration in this patient include infections with CMV, herpes simplex virus, varicella–zoster virus, influenza A or B viruses, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). All these viruses can cause a systemic illness with pneumonitis. However, the radiologic features of pulmonary nodules are not typical of these diagnoses. Furthermore, this patient presents with profound hypotension, which is not suggestive of a viral cause. In addition, the patient is taking valganciclovir, which is highly effective in preventing herpesvirus infections.⁷

Given the patient's history of sarcoidosis, this diagnosis must be considered, since sarcoidosis can cause pulmonary nodules. Cancer is a common cause of pulmonary nodules; however, the patient has no smoking history, which makes lung cancer unlikely. Post-transplantation lymphoproliferative disorder (PTLD) warrants consideration in this patient, although his positive EBV test before transplantation does not support this diagnosis because it occurs more commonly in solid-organ transplant recipients after new infection with EBV.¹ It is possible that the patient has metastatic lesions that are not of pulmonary origin; however, a primary cancer was not identified elsewhere on imaging.

SPLENOMEGALY

The presence of splenomegaly in this patient is an important factor in further narrowing the differential diagnosis of pulmonary nodules. Infections such as tuberculosis and nontuberculous mycobacterial and *H. capsulatum* infections remain possible diagnoses. Among the noninfectious causes of pulmonary nodules that are likely in this patient, only sarcoidosis and PTLD are associated with splenomegaly.

HYPERCALCEMIA

Does the finding of hypercalcemia help to further refine the differential diagnosis? Hypercalcemia has many possible causes, but very few are associated with concurrent pulmonary nodules and splenomegaly. Granulomatous diseases such as sarcoidosis, histoplasmosis, and tuberculosis are

associated with hypercalcemia.^{8–10} The mechanism is postulated to be related to a high serum calcitriol level that may be due to overproduction by granulomas.^{10–12} PTLD has also been associated with hypercalcemia, although less frequently than granulomatous diseases.¹³ This patient's imaging shows no evidence of extrapulmonary lymphadenopathy, which makes PTLD an unlikely diagnosis. After factoring in the finding of hypercalcemia, we are left with three diagnoses with overlapping pathological and clinical features: sarcoidosis, histoplasmosis, and tuberculosis. All three conditions are capable of mimicking each other.

DIFFERENTIATION OF GRANULOMATOUS DISEASES

Sarcoidosis

Sarcoidosis is a multiorgan granulomatous disease that frequently causes pulmonary nodules, mediastinal lymphadenopathy, splenomegaly, and hypercalcemia. This patient has a history of multiorgan sarcoidosis predominantly involving the lung and kidney. Typically, sarcoidosis is treated with immunosuppression.¹⁴ Despite the fact that immunosuppression is administered to prevent kidney allograft rejection, sarcoidosis recurs after kidney transplantation in up to 27% of persons.¹⁵ Recurrence of sarcoidosis after transplantation is rarely fulminant and is often detected on biopsy in asymptomatic patients.¹⁵

Histoplasmosis

Histoplasmosis is caused by infection with *H. capsulatum*. The clinical presentation can range from asymptomatic pulmonary nodules to fulminant disseminated disease with rapidly progressive fever, fatigue, pulmonary nodules, splenomegaly, and hypercalcemia. Disseminated disease is most common in immunosuppressed persons.¹⁶ *H. capsulatum* is a fungus that is endemic worldwide, but it is hyperendemic in the Ohio and Mississippi River valleys.¹⁷ The incidence of diagnosed histoplasmosis in the United States overall is 3.4 cases per 100,000 persons; however, in the northeastern United States, the incidence is only 1.1 cases per 100,000 persons.¹⁸ In New England, where this patient lives, the incidence is very low among persons who have not traveled outside the region.¹⁹ The overall incidence of histoplasmosis after transplantation is approximately 0.1% in the United States.²⁰ Histoplasmosis that develops after transplantation can occur

as a result of new primary infection, donor-derived transmission, or rarely, reactivation of previous infection.¹⁶

Tuberculosis

Tuberculosis is one of the most common infections worldwide and is a major cause of death. Owing to a concerted public health effort over the course of many decades, the incidence in the United States is very low — at approximately 2.2 cases per 100,000 persons — and typically occurs in persons who have resided in a region outside the United States where the disease is endemic.²¹ The post-transplantation incidence overall is between 1.2% and 6.4% in high-income countries.²² The clinical presentation can vary, but among solid-organ transplant recipients (such as this patient), 16% present with disseminated tuberculosis.⁵ The majority of cases will occur within 1 year after transplantation and can be caused by reactivation of latent tuberculosis, newly acquired tuberculosis after transplantation, or donor-derived tuberculosis.²³

In this case, we can rule out sarcoidosis, given that this patient's presentation is fulminant rather than indolent. Despite having a consistent clinical syndrome, the patient has no clear risk factors for either histoplasmosis or tuberculosis; therefore, donor-derived infection must be considered.

DONOR-DERIVED INFECTION

In any unwell solid-organ transplant recipient, the epidemiologic exposures of the recipient must be considered, but so must the exposures of the donor. Donor-derived diseases are an important cause of post-transplantation death and complications.²⁴ Potential organ donors are screened extensively for transmissible diseases through questionnaires (in living donors) and laboratory screening tests, primarily to identify and manage any potential bloodborne infections such as infections with human immunodeficiency virus, hepatitis B virus, and hepatitis C virus. However, other donor-derived diseases, including cancer, can occur, as can other infections such as histoplasmosis and tuberculosis.²⁵ Deceased organ donors are not routinely screened for latent tuberculosis infection or for histoplasmosis because there is usually insufficient time to allow for such screening.

We are provided with no information about the epidemiologic exposures of the donor; the absence of such information is not uncommon in clinical practice. However, there are data to guide clinicians. A 10-year review of the incidence of donor-derived diseases after solid-organ transplantation in the United States showed that tuberculosis accounted for 4% of all donor-derived infections.²⁴ This review also showed that donor-derived tuberculosis occurred up to 148 days after transplantation, whereas donor-derived histoplasmosis occurred within 90 days after transplantation.²⁴

In this patient who had no risk factors for tuberculosis, the clinical syndrome of a progressive, fulminant illness with fever, innumerable pulmonary nodules, splenomegaly, and hypercalcemia occurring 7 months after kidney transplantation is most likely due to donor-derived infection with *M. tuberculosis*. Some patients may have risk factors for tuberculosis that are not identified initially, and once such risk factors are identified, a thorough questioning of the patient is needed. To establish a diagnosis of tuberculosis, I would obtain pulmonary specimens for an acid-fast bacilli smear, nucleic acid amplification testing, and culture. I would begin the investigation by obtaining an induced sputum sample but would rapidly pursue bronchoscopy and biopsy of a pulmonary nodule if acid-fast staining and nucleic acid amplification testing are negative. I would evaluate concurrently for several of the diagnoses discussed previously, given the critical nature of this patient's presentation and the possibility that an immunocompromised host may have multiple coexisting diseases at the time of presentation.

DR. MATTHEW B. ROBERTS' DIAGNOSIS

Mycobacterium tuberculosis infection (most likely donor-derived infection).

PATHOLOGICAL DISCUSSION

Dr. Caroline Hilburn: Bronchoalveolar lavage (BAL) was performed; examination of a smear of the BAL fluid was negative for acid-fast bacilli. Given the clinical suspicion for infection, a sputum specimen was obtained for culture and smear,

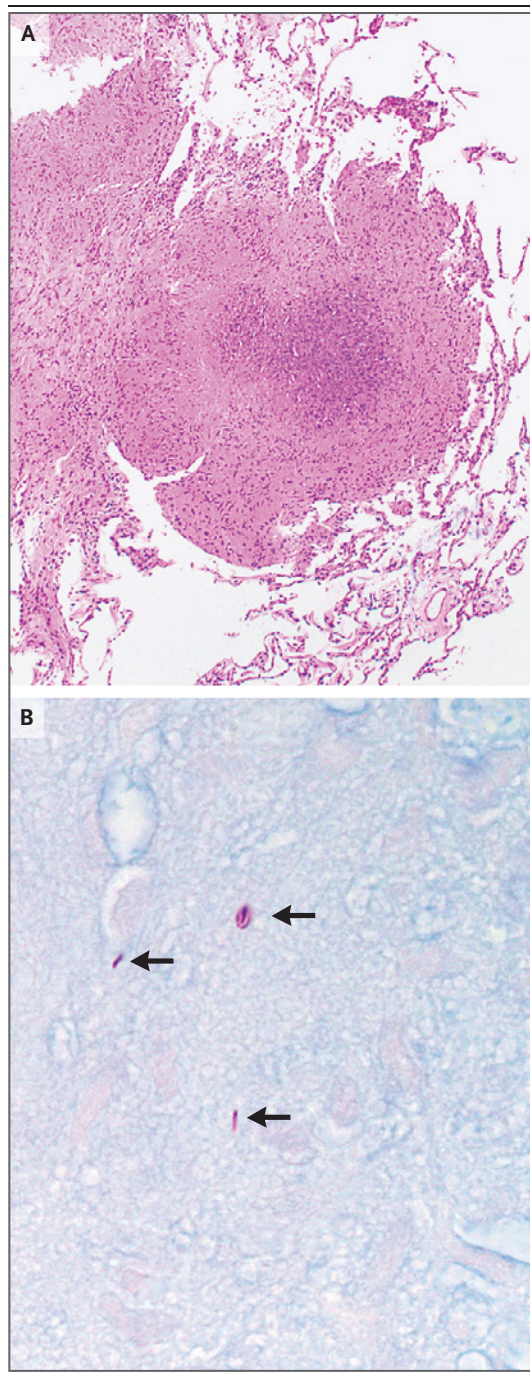


Figure 3. Transbronchial-Biopsy Specimens of the Lung.

Hematoxylin and eosin staining of a right lower lobe specimen (Panel A) shows lung parenchyma with well-demarcated granulomas. The granulomas are characterized by a loose collection of histiocytes with focal palisading around a central region of necrosis that contains nuclear debris. Acid-fast staining (Panel B) shows multiple slender forms within the region of necrosis that are morphologically consistent with *Mycobacterium tuberculosis* complex (arrows).

ulomas (Fig. 3A). The granulomas showed palisading histiocytes forming a ring around a central region of necrosis. Acid-fast staining showed rare slender bacilli with morphologic features that were consistent with *M. tuberculosis* complex (Fig. 3B).

PATHOLOGICAL DIAGNOSIS

Mycobacterium tuberculosis complex infection.

DISCUSSION OF MANAGEMENT

Dr. Arthur Y. Kim: In a recipient of a solid-organ transplant, multiple parts of the immune system are modulated to allow survival of the graft; these include T cells and macrophages, both of which are important in the control of tuberculosis.²⁶ Reduction of immunosuppression is a standard practice used in transplant recipients who present with severe infections in order to maximize immune responses as an adjunct to specific anti-infective therapy. Once the diagnosis of tuberculosis was established in this patient, treatment with mycophenolate mofetil was discontinued. Antituberculosis therapy was initiated with a standard regimen of isoniazid, rifampin, pyrazinamide, and ethambutol, since we had no reason to suspect primary drug resistance. The use of standard combination therapy for tuberculosis is often challenging in transplant recipients owing to additive toxic effects from multiple medications and drug interactions; however, this patient was not receiving tacrolimus or sirolimus, which allowed for the use of a rifampin-containing regimen.

We also considered the source of the patient's infection. Could this be reactivation of latent tuberculosis? The patient's pretransplantation workup included a negative interferon- γ release

and both were positive for acid-fast bacilli. Confirmatory nucleic acid testing of the sputum specimen detected *M. tuberculosis* complex.

In addition, a transbronchial-biopsy specimen was obtained from the right lower lobe. The specimen consisted of fragments of benign bronchial and alveolar tissue with multiple necrotizing gran-

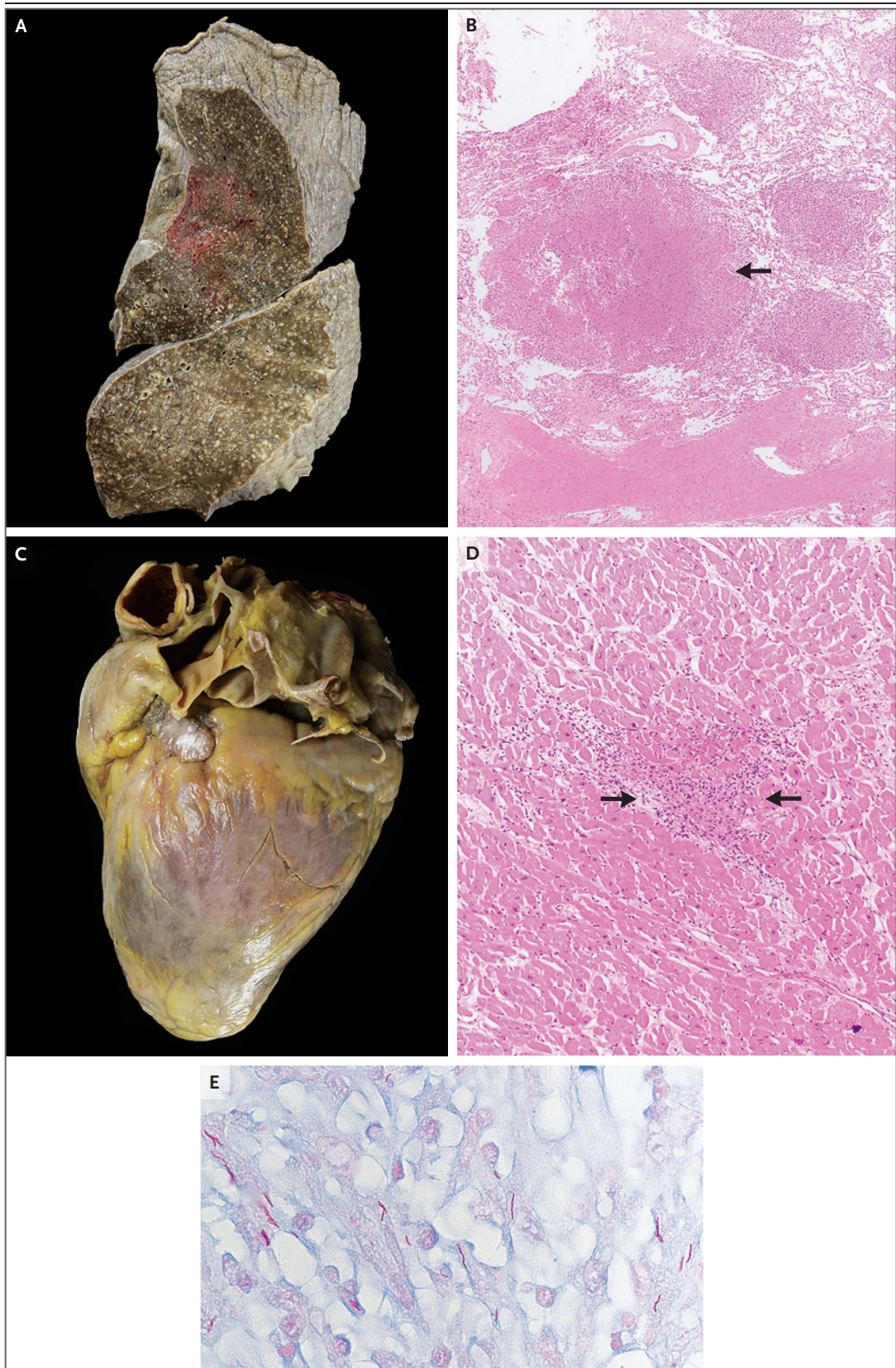


Figure 4 (facing page). Autopsy Specimens.

Gross examination of the lungs (Panel A) reveals the presence of dispersed, punctate lesions with a resemblance to millet seeds throughout both lungs. Hematoxylin and eosin staining (Panel B) shows that the lesions are necrotizing granulomas (arrow). Gross examination of the heart (Panel C) reveals a normal pericardium, but hematoxylin and eosin staining of an examined section (Panel D) shows minimal granulomatous inflammation (arrows). Necrotizing granulomas are present in multiple organs, including the mesenteric adipose tissue, with acid-fast staining (Panel E) confirming the presence of acid-fast bacilli.

assay, but this test can be falsely negative, especially in patients with end-stage organ disease.²⁷ Although the prevalence of tuberculosis is declining in the United States, among U.S.-born persons, the prevalence is higher among patients receiving hemodialysis.^{28,29} In this patient, acquisition of tuberculosis during the post-transplantation period was not considered to be likely because the patient had very few contacts outside medical visits, and follow-up information obtained from these contacts was unrevealing.

Could this case of tuberculosis be donor-derived? The limited available information regarding the donor did not suggest risk factors for tuberculosis. A presentation that would have suggested donor-derived infection would be a greater burden of disease at or near the transplanted graft, such as a chronic or fistulous wound infection. Although a culture of urine obtained from the patient ultimately grew *M. tuberculosis*, such a finding may also be due to disseminated disease and is not specific for donor-derived infection. The development of tuberculosis in other persons who received organs from this donor would also be highly suggestive of donor-derived infection.³⁰ However, no clinical or laboratory evidence suggested a diagnosis of tuberculosis in the other recipient of an organ from the same donor. Molecular techniques can complement traditional contact tracing and can sometimes shed light on the networks and directionality of transmission.³¹ Ultimately, several lines of investigation were unable to prove the origin of this patient's tuberculosis. Out of an abundance of caution, despite a lack of clinical or laboratory evidence of tuberculosis, chemoprophylaxis with isoniazid was administered to the other recipient who received an organ from this donor.

HOSPITAL COURSE

Dr. Lee: Although tuberculosis that results in multi-organ failure (also known as Landouzy septice-mia) is a rare cause of sepsis, it has been well documented. In addition to treatment with isoniazid, rifampin, pyrazinamide, and ethambutol, the patient received intravenous ganciclovir for an elevated CMV viral load of 9340 IU per milliliter of blood. Ultimately, culture from the transbronchial-biopsy specimen grew *Pseudomonas aeruginosa*, so treatment with cefepime was continued.

Throughout this patient's hospital course, glucocorticoids were administered for the treatment of hypotension and septic shock, since treatment with glucocorticoids is associated with rapid reversal of shock; however, its effect on survival is unclear.³²⁻³⁶ The administration of glucocorticoids seems to benefit patients with disseminated tuberculosis, especially those with tuberculosis meningitis, moderate-to-severe immune reconstitution inflammatory syndrome, and possibly tuberculosis-related respiratory failure.³⁷⁻³⁹

Because of this patient's persistent septic shock, he began to have worsening acute kidney injury for which hemodialysis was initiated. Unfortunately, on hospital day 6, new diffuse ST-segment elevations were seen on electrocardiography. Echocardiography revealed new biventricular dysfunction, with a left ventricular ejection fraction of 16% — a decrease from a level of 70% obtained 3 days earlier. On the basis of these findings, new cardiomyopathy was diagnosed. Shortness of breath developed, and ultimately, the patient's trachea was intubated and mechanical ventilation was initiated. Norepinephrine was started to maintain his blood pressure.

Dr. Erin E. Coglianese: Coronary angiography performed on the same day revealed no epicardial coronary artery disease. In the absence of coronary artery disease, the differential diagnosis for acute cardiomyopathy included stress cardiomyopathy, CMV myocarditis, and lymphocytic myocarditis; however, sepsis-induced cardiomyopathy mediated by pathogen-associated molecular patterns, damage-associated molecular patterns, and cytokines was considered to be the most likely diagnosis.⁴⁰ Tuberculosis-associated myocarditis is rarely described (0.1 to 2% incidence) and more frequently involves the pericardium.⁴¹

Progressive cardiogenic shock developed in

the patient. A percutaneous left ventricular assist device cannula was placed through the right femoral artery to support cardiac output. Despite this intervention, the patient died on hospital day 6.

AUTOPSY

Dr. Hilburn: An autopsy was performed at the request of the family. Gross examination of the lungs (Fig. 4A) was notable for numerous punctate lesions dispersed throughout all the lobes. The lesions were small, round, and tan and were well-demarcated from the surrounding parenchyma. Histologic examination revealed necrotizing granulomas with palisading histiocytes around a core of necrosis. Gross examination of the heart revealed a normal pericardium with a smooth, glistening epicardium. There was no interstitial or replacement fibrosis that could indicate a previous ischemic event. The wall thickness and valves were normal. The coronary arteries had minimal calcific atherosclerosis, with the

most substantial focus being 50% stenosis in the left main coronary artery. On histologic examination of the heart, there was minimal involvement by granulomatous inflammation, with a single poorly formed granuloma. There were no morphologic diagnostic features of myocarditis. Examination of the kidney allograft identified rare necrotizing granulomas. A single acid-fast bacillus was identified in the kidney allograft. In addition, necrotizing granulomas were identified in the mesenteric adipose tissue (Fig. 4B), spleen, liver, periaortic lymph nodes, testis, bladder, and thyroid. Miliary *M. tuberculosis* with hematogenous spread was reported as the underlying cause of death.

FINAL DIAGNOSIS

Disseminated *Mycobacterium tuberculosis* infection.

This case was presented at the Medicine Case Conference. Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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